Chapter 2a – Rational Expressions – REVIEW
MCR3U
Jensen

Section 1: Negative and Rational Exponents

1) Evaluate. Express as a fraction in lowest terms.

a) 10^{-1}

b) 4^{-2}

c) $3^{-2} + 9^{-1}$

d) $5^{-3} + 5^{0}$

e) $\left(\frac{1}{5}\right)^{-1}$

f) $\left(\frac{3}{4}\right)^{-3}$

2) Simplify. Express your answers using only positive exponents.

a) $(x^{-2})(x^{-1})(x^{0})$

b) $(3km^{2})(2k^{-2}m^{-2})$

c) $w^{-3} \div w^{-2}$

d) $\frac{u^{-2}v^{3}}{u^{-3}v^{-2}}$

e) $(z^{-3})^{-2}$

f) $(2ab^{-1})^{-2}$

3) Simplify. Express your answers using only positive exponents.

a) $(4a^{-2})(-2a^{-3})$

b) $\frac{(2x^2y)^{-2}(3xy)^{-1}}{(6x^2y^2)^{-2}}$

c) $\left(\frac{1}{4x^2}\right)^{-2}$

d) $\left(\frac{6a^3}{4b^4}\right)^{-2}$

4) Evaluate.

a) $\sqrt[3]{64}$

b) $\sqrt[4]{625}$

c) $\sqrt[5]{-3125}$

d) $\left(\frac{1}{64}\right)^{\frac{1}{6}}$

e) $27^{\frac{2}{3}}$

f) $(-1000)^{\frac{4}{3}}$

g) -4^{-3}

h) $\left(\frac{3}{4}\right)^{-2}$

i) $\left(-\frac{27}{125}\right)^{-\frac{2}{3}}$

5) Simplify. Express answers using only positive exponents.

a) $n^{\frac{1}{2}} \times n^{\frac{1}{3}} \times n^{\frac{1}{4}}$

b) $(27y^3)^{\frac{1}{3}} \times \left(\frac{1}{16y^4}\right)^{-\frac{3}{4}}$

c) $(27x^6)^{\frac{2}{3}} \div (9x^4)^{\frac{1}{2}}$

d) $\frac{x^{-\frac{2}{3}}}{x^{-\frac{4}{5}}}$

Section 2: Rational Expressions

6) Simplify each expression and state all restrictions on x.

a) $\frac{x+7}{x^2+10+21}$

b) $\frac{x^2-64}{x-8}$

c) $\frac{x^2-9}{x^2-8x+15}$

7) Simplify each expression and state the restrictions.

a) $\frac{3x^2}{5xy} \times \frac{20xy^3}{12xy}$

b) $\frac{150a^3b^4}{20a^2b} \div \frac{6b}{8ab^2}$

c) $\frac{1}{3x} + \frac{5}{2x^2}$

d) $\frac{4}{x-6} - \frac{3}{x-4}$

8) Simplify each expression and state restrictions.

a) $\frac{x^2+7x}{3x+21} \times \frac{x^2+3x+2}{x+2}$

b) $\frac{x^2+4x-60}{3x+30} \div \frac{x^2-8x+12}{6x-12}$

c) $\frac{3}{x^2+7x+10} - \frac{5x}{x^2-4}$

d) $\frac{-10x}{x^2+18x+32} + \frac{12x}{x^2+6x-160}$

9) Simplify each expression and state any restrictions

a) $\frac{x-8}{x+7} \times \frac{x+15}{x^2+12x-45}$

b) $\frac{x^2+12x+20}{x+5} \div \frac{x^2+7x-30}{x+10}$

c) $\frac{x+3}{x-7} - \frac{x+9}{x-2}$

d) $\frac{x+8}{x+3} + \frac{x-6}{x^2+9x+18}$

e) $\frac{5x+1}{2x-1} - \frac{3x-3}{1-2x}$

Answers

1) a) $\frac{1}{10}$ **b)** $\frac{1}{16}$ **c)** $\frac{2}{9}$ **d)** $\frac{126}{125}$ **e)** 5 **f)** $\frac{64}{27}$

2) a) $\frac{1}{x^3}$ **b)** $\frac{6}{k}$ **c)** $\frac{1}{w}$ **d)** uv^5 **e)** z^6 **f)** $\frac{b^2}{4a^2}$

3) a) $-\frac{8}{a^5}$ **b)** $\frac{3y}{x}$ **c)** $16x^4$ **d)** $\frac{4b^8}{9a^6}$

4) a) 4 **b)** 5 **c)** -5 **d)** $\frac{1}{2}$ **e)** 9 **f)** $10\,000$ **g)** $-\frac{1}{64}$ **h)** $\frac{16}{9}$ **i)** $\frac{25}{9}$

5) a) $n^{\frac{13}{12}}$ **b)** $24y^4$ **c)** $3x^2$ **d)** $x^{\frac{2}{15}}$

6) a) $\frac{1}{x+3}$, $x \neq -7, x \neq -3$ **b)** $x + 8$, $x \neq 8$ **c)** $\frac{x+3}{x-5}$, $x \neq 3, 5$

7) a) xy , $x \neq 0, y \neq 0$ **b)** $10a^2b^4$, $a \neq 0, b \neq 0$ **c)** $\frac{2x+15}{6x^2}$, $x \neq 0$ **d)** $\frac{x+2}{(x-4)(x-6)}$, $x \neq 4, x \neq 6$

8) a) $\frac{x(x+1)}{3}$, $x \neq -7, -2$ **b)** $2, x \neq -10, 2, 6$ **c)** $\frac{-5x^2-22x-6}{(x+5)(x+2)(x-2)}$, $x \neq -5, -2, 2$

d) $\frac{2x(x+62)}{(x+16)(x+2)(x-10)}$, $x \neq -16, -2, 10$

9) a) $\frac{x-8}{(x+7(x-3)}$, $x \neq -15, -7, 3$ **b)** $\frac{(x+10)(x+2)}{(x+5)(x-3)}$, $x \neq -10, -5, 3$ **c)** $\frac{-x+57}{(x-7)(x-2)}$, $x \neq 2, 7$

d) $\frac{x^2+15x+42}{(x+6)(x+3)}$, $x \neq -6, -3$ **e)** $\frac{8x-2}{2x-1}$, $x \neq \frac{1}{2}$